



**Dissertation Sentiment Analysis on Social Media Posts Multilingual /
Code-Mixed Sentiment**

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ABSTRACT

Sentiment analysis of industry-focused social media data in Roman Urdu poses a very difficult computational task. The reasons for this, really amount to three properties--the lack of unified spelling in the language, the widespread and extended use in text of English with Roman Urdu, and the out-of-vocabulary failures that such properties bring about in fixed-vocabulary models. In the form of a straightforward story, the GhostNet model displays its own subtle language remnants of this network origin. The Facts: If commentators wish to clarify whether or not edge villages without linesmen are being served, they should write this woman a letter stating that "we wait for service." In contrast to Unix, which is extensively utilised by research labs and telecom businesses, Linux is not well-known in China's major telecom industry and is hardly utilised by the nation's most esteemed colleges. Last year, when dining in Beijing with some American friends, I expressed my opinion that the trappings of power didn't have the same effect on the typical American as they did in China. They expressed many things I could observe and agreed with me. Additionally, these oral antecedents are probably going to provide first-hand information for lexicography. Both goals are pursued by the study. Regarding the Roman Urdu Sentiment Analysis Dataset, which Google produced in two Because XLM-RoBERTa's subword tokenisation resolves the out-of-vocabulary issue that besets fixed-vocabulary sequential models, the improvement is most noticeable in the positive class, where it decreased the false negative rate from 21.3% to 16.1%. The refined model also demonstrated 81% accuracy and a weighted F1-score of 0.80 in a cross-domain stress test on the Roman Urdu-English Code-Switched Emotion Dataset. The majority of misclassifications involve mood conveyed through irony and culturally distinctive indirect language, according to error analysis, showing a constraint that cannot be addressed by architectural changes alone. As demonstrated by our experiment, these results support the notion that multilingual transformer systems provide significant advantages to low-resource, orthographically inconsistent, code-mixed languages like Roman Urdu. In the end, the creation of new models will be just as important to this field's advancement as deeper annotation systems.

Section 1- Introduction

There is a higher proportion of people who have been using social media and have access to smartphones. Everything changed after YouTube, Facebook, the first launch of an iPhone and Android phones — even older people now know how to watch something on YouTube and send something on WhatsApp to their family and friends. Companies that have been providing customer services and improving day by day want to know what people are saying, but sometimes they cannot understand it.

Roman Urdu is a combination of two languages — English and Urdu. Most of the time, people from Asian countries like Pakistan and other South Asian countries express their opinions on social media in their own language, but with English words added into the sentences or phrases.

On top of that, there is no fixed spelling, constant code-mixing with English, and complex grammar, which makes it very difficult for machine learning models to catch and understand what is being said. These are structural problems within the communication itself, not small technical issues that can be easily solved. The research has been conducted by leading architectures in this field. With minimal fine-tuning, rank first out of 28 competing teams in a shared task focused on detecting subjective emotional tone in code-mixed text.

Despite much research, there has been no direct comparison of LSTM and XLM-RoBERTa to determine which model outperforms the other, a critical inquiry for practical applications of these models. These findings address both gaps by conducting a controlled experiment and provide three primary contributions: a direct comparison of Bi-LSTM and fine-tuned XLM-RoBERTa in Roman Urdu sentiment classification, demonstrating the improved model's performance on an entirely different dataset on which it was not previously trained, along with an error analysis highlighting the models' inaccuracies, elucidating the reasons for these errors by incorporating insights into the practical usage of Roman Urdu. Impacts and contributions like these make an understanding of the state and deficiencies in Roman Urdu sentiment analysis.

SECTION 2 — LITERATURE REVIEW

Since its inception in natural language processing, sentiment analysis has undergone substantial development. Early strategies focused on conventional machine learning techniques like logistic regression, support vector machines, and naive bayes. Bag-of-words representations, which treat text as a collection of individual words without taking into account their order, are commonly employed in these techniques. Although this method allowed for early sentiment classification, it disregarded linguistic structure, which frequently conveys meaning.

Sentences like "the food was good" and "the food was not good," for example, use identical words yet convey entirely distinct ideas. Because bag-of-words models ignore word order and relationships, they are unable to account for this disparity. In languages like Roman Urdu, where structure and diversity are crucial, this restriction is much more noticeable.

Among the first to use LSTM models for Roman Urdu sentiment analysis were Ghulam et al. (2019), who demonstrated definite advancements over conventional techniques. Their research showed that performance improves when sequence and context are understood. Nevertheless, their method continued to rely on set vocabularies, which created problems when handling spelling changes.

Khan et al. (2022) combined CNN and LSTM architectures to further enhance performance. Both long-range dependencies and local patterns were captured by their hybrid approach. Additionally, they examined several embedding methods and discovered that Word2Vec outperformed more

intricate models such as BERT for Roman Urdu. This brought to light a crucial problem: sophisticated models frequently need huge, reliable datasets, which aren't always accessible for low-resource languages.

In order to enhance performance, Li et al. (2022) investigated transfer learning and introduced attention mechanisms. In order to improve sentiment categorisation, their model concentrated on determining the most significant terms in a sentence. Although their findings were impressive, the model's complexity greatly increased, making it more difficult to replicate and use in real-world situations.

The employment of multilingual transformer models, such as XLM-RoBERTa, is a more recent development. In contrast to previous methods, these models employ subword tokenisation to accommodate unfamiliar or inconsistent terms and are pre-trained on big multilingual datasets. In code-mixed language challenges, where various languages are utilised together in informal text, Ou and Li (2020) showed how effective this strategy is.

This is a fact that Roman Urdu, combines parts of both Urdu and English with erratic spelling. Instead of attempting to get around these issues, transformer models tackle them structurally.

All things considered, the area has gone from straightforward statistical techniques to sophisticated deep learning and transformer-based methods. There is still a noticeable gap in spite of these advancements. Bidirectional LSTM and XLM-RoBERTa have not been directly compared on the RUSAD dataset under the same conditions, nor has their performance in other domains been thoroughly assessed. Both of these gaps are intended to be filled by this study.

SECTION 3 — METHODOLOGY

This study employs a comparative experimental methodology to assess two deep learning models for binary sentiment categorisation on Roman Urdu social media text: a refined XLM-RoBERTa and a bidirectional LSTM. The entire process is divided into four primary phases: choosing and preparing the dataset, employing data pre-processing methods, creating and training the models, and assessing their effectiveness using common metrics.

3.1 Datasets

Primary Dataset — RUSAD

11,000 user reviews that were gathered from Pakistani social media sites and labelled with binary emotion labels—positive and negative—make up RUSAD. The dataset, which was obtained from the UCI Machine Learning Repository and used in earlier published studies such as Li et al. (2022) and Khan et al. (2022), is a recognised standard in Roman Urdu

sentiment analysis research. This benchmark status is important from a methodological standpoint since it allows the results of this study to be directly compared with published performance data, offering a more robust foundation for assessing model performance than would be possible with an unvalidated corpus.

Recognising that RUSAD uses a binary annotation scheme (positive and negative) without a neutral class is crucial. Section 5 discusses the consequences for model performance and generalisability of this intentional dataset design constraint rather than an oversight of this investigation.

A final corpus of 10,999 cases with a nearly balanced class distribution of 5,699 positive and 5,300 negative evaluations was obtained after one aberrant occurrence with a null label was found during exploratory analysis and eliminated prior to modelling.

Stress-Test Dataset — Roman Urdu-English Code-Switched Emotion Dataset

A second dataset was used only for out-of-domain stress testing in order to assess cross-domain robustness. Twenty thousand tweets are annotated with six emotion categories in the Roman Urdu-English Code-Switched Emotion Dataset: Happy, Anger, Sad, Fear, Surprise, and Neutral.

Sentiment mapping was used and positive sentiment was associated with instances of happiness, whereas negative sentiment was attached with anger, sadness, and fear. Because their emotion polarity is intrinsically ambiguous and cannot be consistently transferred to a binary schema without introducing systematic noise, surprise and neutral occurrences were eliminated. This produced 8,147 instances in the stress-test corpus. The stress-test dataset was only used to assess the fine-tuned XLM-RoBERTa mode--m's capacity for generalisation; it was never utilised for training.

The original label distribution of the emotion dataset prior to sentiment mapping is presented in Table 7. Neutral and Surprise instances were excluded on the grounds that their sentiment polarity is ambiguous and cannot be reliably mapped to a binary

Emotion	Count	Mapped To
Neutral	11,629	Excluded
Happy	3,720	Positive
Anger	3,487	Negative
Sad	712	Negative
Fear	228	Negative
Surprise	224	Excluded
Total Used	8,147	—

schema without introducing systematic noise

Table 7: Emotion label distribution in the Roman Urdu-English Code-Switched Emotion Dataset prior to sentiment mapping.

3.2 Data Pre-processing

There is a lot of noise in Roman Urdu social media material that, if ignored, would provide unrelated signals to model training. Before tokenisation, the RUSAD corpus was subjected to a uniform

pre-processing workflow. Regular expressions were used to eliminate URLs, @mentions, and hashtags because, in the context of this work, they don't convey any sentiment. To lessen orthographic variability, all text was changed to lowercase and special letters and punctuation were removed. To eliminate superfluous spacing brought about by the cleaning procedure, whitespace normalisation was used. The re library was used to create this pipeline in Python, and it was applied consistently to all 10,999 instances.

The pre-processing pipeline was purposefully conservative. Aggressive normalisation runs the risk of eliminating significant tokens from Roman Urdu text, which contains acceptable alphanumeric mixed content, such as numerals contained within phrases. As a result, the pipeline eliminated structural noise while maintaining word-character sequences.

3.3 Model Architectures

Model 1 — Bidirectional LSTM (Baseline)

Because of its proven ability to handle sequential text input, the Bidirectional LSTM model was chosen as a baseline. Conventional LSTM models can't completely comprehend context because they only process text in one direction. Meaning in sentiment analysis frequently depends on a sentence's previous and later words. For instance, to comprehend the negative feeling in a statement like "the food was not good at all," one must take into account both the word "not" and the subsequent phrase.

By analysing the sequence in both forward and backward orientations, bidirectional LSTM gets over this restriction. As a result, the model is able to comprehend the sentence more fully by capturing context from both sides of a word.

This model was chosen on purpose. Bi-LSTM performs effectively and serves as a robust baseline rather than a weak comparison point, according to earlier research in Roman Urdu sentiment analysis. Any improvement seen with more sophisticated models is certain to be significant and trustworthy when a strong baseline is used.

The model's design starts with an embedding layer that uses a vocabulary size of 10,000 and an embedding dimension of 128 to translate text into numerical representations. In order to balance coverage and computational efficiency, this size was selected. After that, a bidirectional LSTM layer with 64 units in each direction generates combined contextual characteristics. To lessen overfitting, dropout layers with a rate of 0.5 are used. The final sigmoid output layer, which carries out binary classification, is preceded by a dense layer with 64 units and ReLU activation.

The Keras Tokeniser is used to tokenise text data, with a maximum sequence length of 100 tokens. The Adam optimiser and binary cross-entropy loss are used to train the model. By restoring the optimal model weights based on validation results, early stopping is used to avoid overfitting.

Layer	Output Shape	Parameters
Embedding	(None, 100, 128)	1,280,000
Bidirectional LSTM	(None, 128)	98,816
Dropout (0.5)	(None, 128)	0
Dense (ReLU)	(None, 64)	8,256
Dropout (0.5)	(None, 64)	0
Output Dense (Sigmoid)	(None, 1)	65
Total Parameters	—	1,387,137

Table 6: Bi-LSTM model architecture and parameter count.

Model 2 — XLM-RoBERTa (Primary Model)

Because XLM-RoBERTa-base can handle multilingual and highly changeable text, which is directly related to the characteristics of Roman Urdu, it was chosen as the main model. Its extensive pre-training on multilingual data is one of its main advantages. The program can recognise patterns that cut across linguistic boundaries because it was trained on an enormous dataset that covered 100 languages.

This is particularly helpful for Roman Urdu, as users often combine English and Urdu in one sentence. XLM-RoBERTa already has some knowledge of both linguistic systems and their interactions, in contrast to models trained on a single language.

This model's tokenisation strategy is another significant benefit. Because Roman Urdu does not adhere to normal spelling norms, a word may exist in more than one form. Due to their reliance on fixed vocabulary, traditional models have difficulty with such variation. Subword tokenisation, which divides unfamiliar words into smaller, more comprehensible components, is how XLM-RoBERTa does this. Instead of considering spelling differences as unknown tokens, this enables the model to handle them more skilfully.

A classification layer was added on top of the pre-trained XLM-RoBERTa model to refine it for this investigation. Because the entire model was trained on the dataset from beginning to end, it was able to adjust to the unique features of Roman Urdu text. To guarantee steady training, a linear scheduler, warmup steps, and a learning rate of 1e-5 were employed. After preliminary tests revealed that larger learning rates produced unstable results, this configuration was selected.

A batch size of sixteen was used for five epochs of training. GPU resources made available by the university's computing infrastructure were used for all tests.

"All experiments were conducted on a Tesla P100-SXM2 GPU with 16GB memory, accessed via the University of Hull High Performance Computing cluster running Python 3.11.10. All random seeds were fixed at 42 across Python, NumPy, TensorFlow, and PyTorch to ensure full reproducibility of results."

3.4 Experimental Design and Evaluation

The model's performance was evaluated using the accuracy, precision, recall, and weighted F1-score. Weighted F1-score was selected as the primary assessment metric because it takes into consideration class imbalance by weighting each class's F1-score by its support. Weighted F1-score offers a more meaningful measure of overall model performance than accuracy alone for datasets with unequal class sizes. Confusion matrices were developed for both models to enable a more thorough examination of misclassification patterns.

The refined XLM-RoBERTa model was tested immediately on the 8,147-instance emotion corpus for the out-of-domain stress test without further fine-tuning, offering a clear indicator of cross-domain generalisation ability. In order to find systemic error patterns, misclassified occurrences were saved and put through qualitative linguistic analysis.

SECTION 4 — RESULTS

The findings are presented in four stages: a preliminary analysis of the dataset before modelling, an in-domain assessment of both architectures on the RUSAD holdout set, an out-of-domain stress test of the refined XLM-RoBERTa model, and a qualitative error analysis of cases that were incorrectly categorised. The results are presented in this section as observed; Section 5 will provide interpretation and theoretical justification.

4.1 Exploratory Data Analysis

After the sole null-label occurrence found during pre-processing was eliminated, the cleaned RUSAD corpus consisted of 10,999 reviews divided into two classes: 5,699 positive and 5,300 negative, accounting for 51.8% and 48.2% of the corpus, respectively. Since a naive majority-class classifier would only achieve slightly above 50% accuracy, it is important to establish the near-balanced class distribution at the outset. This provides a meaningful lower bound against which the reported model performances of 80% and 82% should be understood. These outcomes are not easily attained; rather, they show a real and significant improvement over the random prediction baseline.

A pattern that is typical of social media data is seen in the distribution of review text lengths. The bulk of reviews are less than 200 characters long, and there is a noticeable right skew caused by a smaller number of significantly lengthier texts that are more than 1,000 characters long. The vast majority of examples could be represented without severe truncation using the maximum sequence length of 100 tokens. Figures 1 and 2 show the distribution of class and text length, respectively.

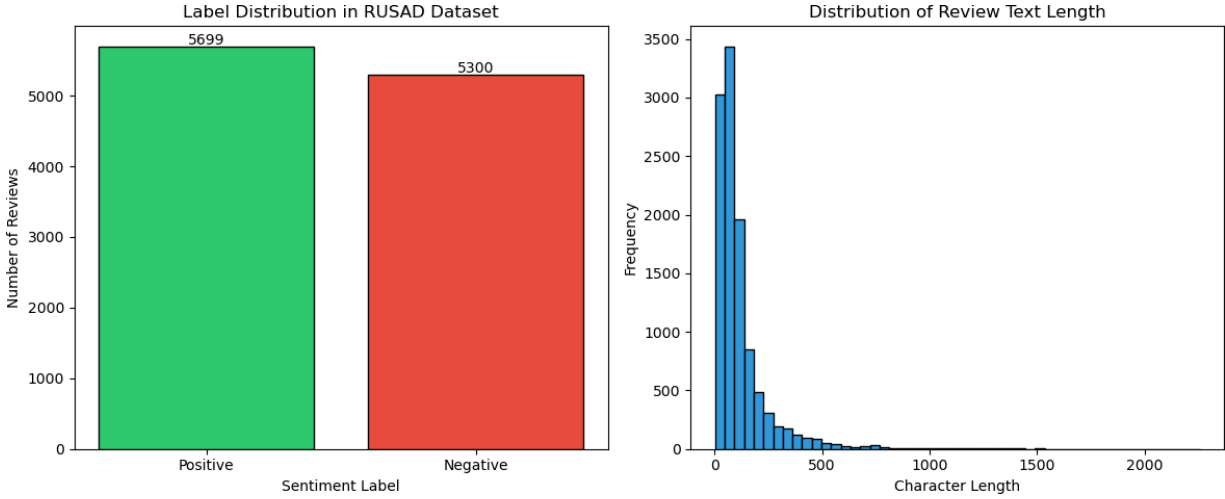


Figure 1: Label distribution and review text length distribution in the RUSAD dataset ($n = 10,999$).

4.2 In-Domain Evaluation — Bi-LSTM

The Bi-LSTM baseline achieved 80% accuracy on the 2,200-instance holdout test set. The full classification report is presented in Table 1.

Class	Precision	Recall	F1-Score	Support
Negative	0.78	0.81	0.80	1,060
Positive	0.82	0.79	0.80	1,140
Weighted Average	0.80	0.80	0.80	2,200

Table 1: Bi-LSTM classification report on the RUSAD holdout test set.

A more detailed view of the areas where the model's errors are concentrated can be found in the confusion matrix shown in Figure 3. There was an error rate of 18.7% on the negative class, with 862 of the 1,060 negative instances in the test set being correctly classified and 198 being inaccurately projected as positive. There was a 21.3% error rate on the positive class, with 897 of the 1,140 positive cases correctly classified and 243 mistakenly projected as negative. As a result, the model is somewhat more likely to incorrectly classify a positive review as negative than a negative review as positive. This trend becomes clearer when compared to the XLM-RoBERTa results in Section 4.3.

A pattern that is both predicted and diagnostically useful can be seen in the training history shown in Figure 4. While validation accuracy plateaued at about 80% starting in the second epoch and did not improve thereafter, training accuracy increased significantly across epochs,

reaching about 97% by the time early halting intervened. Overfitting is clearly identified by the associated difference between training loss, which kept declining, and validation loss, which started to increase after Epoch 1. The model weights from Epoch 1, when validation loss was at its lowest value of 0.4521, were restored by early stopping with a patience of three epochs. This discrepancy between training and validation performance is not a coincidental result of the experimental design.

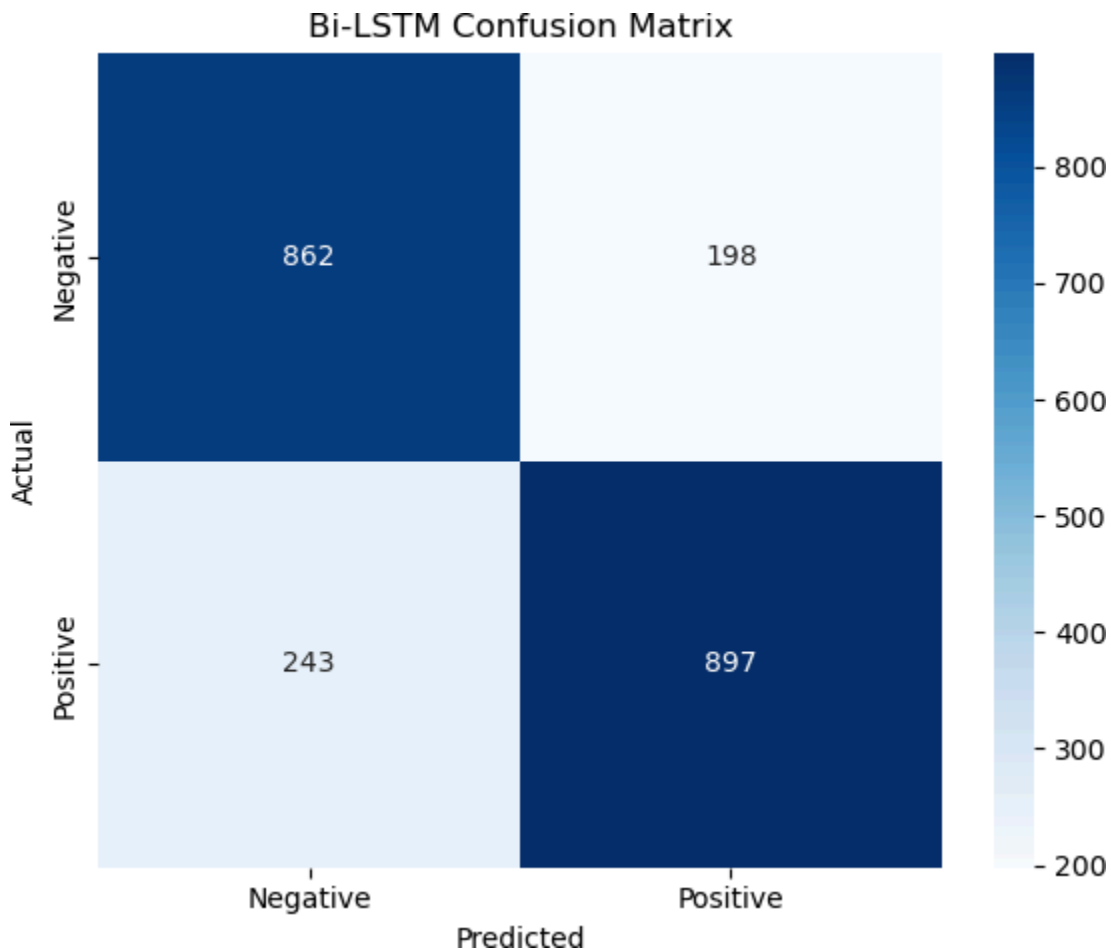


Figure 2: Confusion matrix for the Bi-LSTM baseline model on the RUSAD holdout test set.

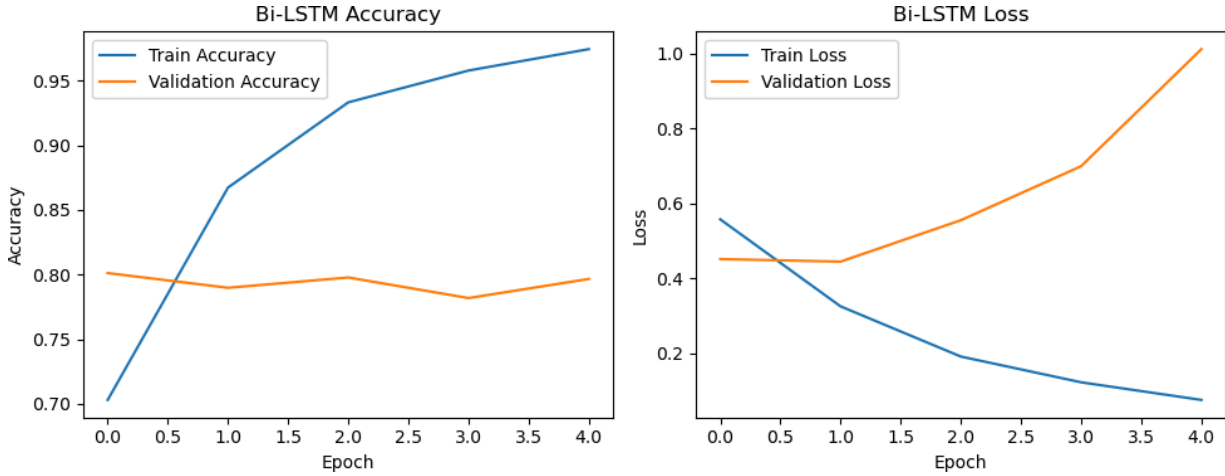


Figure 3: Bi-LSTM training accuracy and loss curves, illustrating the divergence between training and validation performance characteristic of overfitting.

4.3 In-Domain Evaluation — XLM-RoBERTa

On the same holdout test set, refined XLM-RoBERTa achieved 82% accuracy. Table 2 displays the whole classification report.

Class	Precision	Recall	F1-Score	Support
Negative	0.82	0.81	0.82	1,060
Positive	0.82	0.84	0.83	1,140
Weighted Average	0.82	0.82	0.82	2,200

Table 2: XLM-RoBERTa classification report on the RUSAD holdout test set.

Figure 5's confusion matrix reveals that although 956 positive instances were accurately classified and 184 wrongly forecasted as negative, 857 negative examples were correctly classified and 203 incorrectly anticipated as positive. Even while the overall accuracy improvement of two percentage points over Bi-LSTM is small in absolute terms, it is analytically significant since it is distributed unevenly across the two classes. Compared to Bi-LSTM, XLM-RoBERTa properly identified 59 more positive occurrences, lowering the false negative rate on the positive class from 21.3% to 16.1%. The two models' performance on the negative class was largely comparable; Bi-LSTM's F1-score was 0.80, while XLM-RoBERTa's was 0.82. This improvement's asymmetry, which focused on the positive class instead.

Metric	Bi-LSTM	XLM-RoBERTa
Accuracy	80%	82%

F1 Negative	0.80	0.82
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F1 Positive	0.80	0.83
F1 Weighted	0.80	0.82

Table 3: Head-to-head comparison of Bi-LSTM and XLM-RoBERTa on the RUSAD holdout test set.

The Bi-LSTM training history and the training loss curve shown in Figure 6 have quite distinct profiles. From 0.585 in Epoch 1 to 0.277 in Epoch 5, the loss fell steadily and smoothly over all five epochs, showing no indication of the divergence between training and validation performance that defined the Bi-LSTM run. This consistent convergence is indicative of a basic distinction between the two models' initial conditions, which is theoretically explored in Section 5.

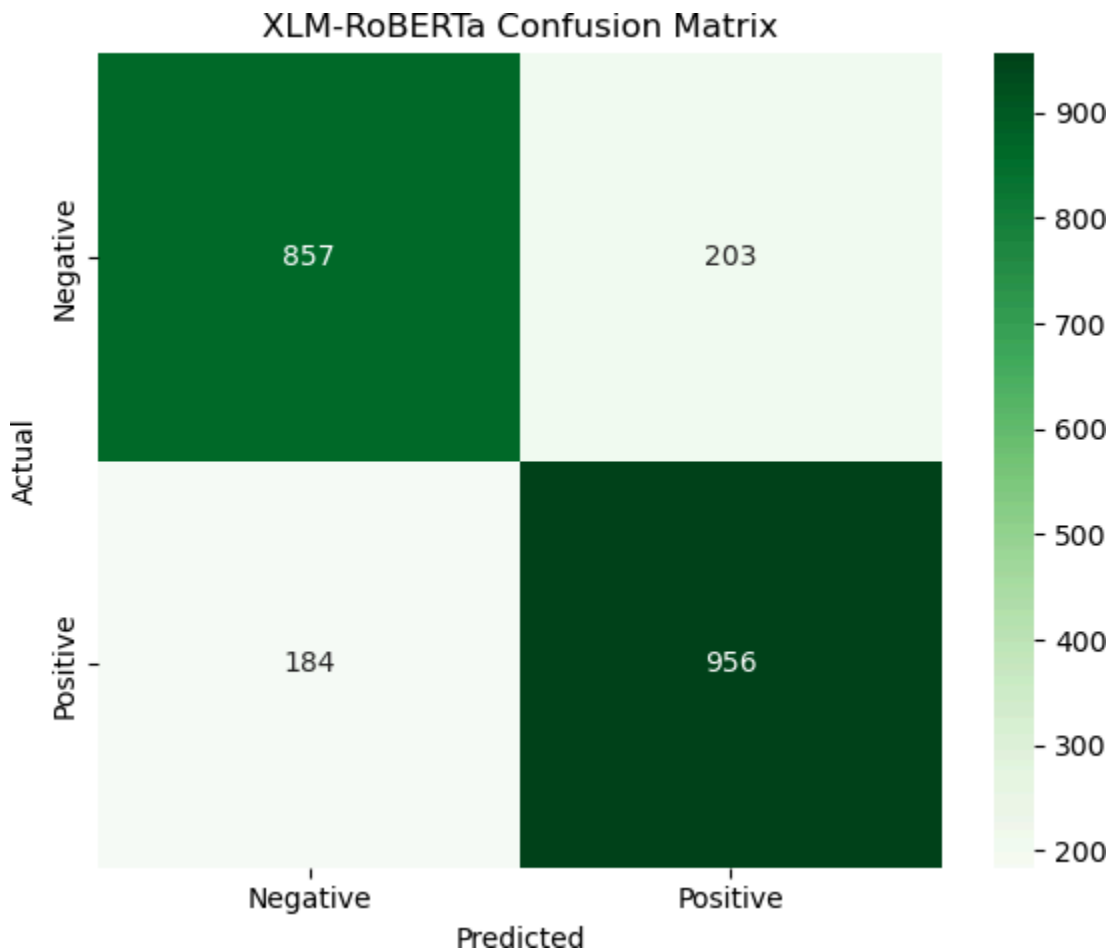


Figure 4: Confusion matrix for the fine-tuned XLM-RoBERTa model on the RUSAD

holdout test set.

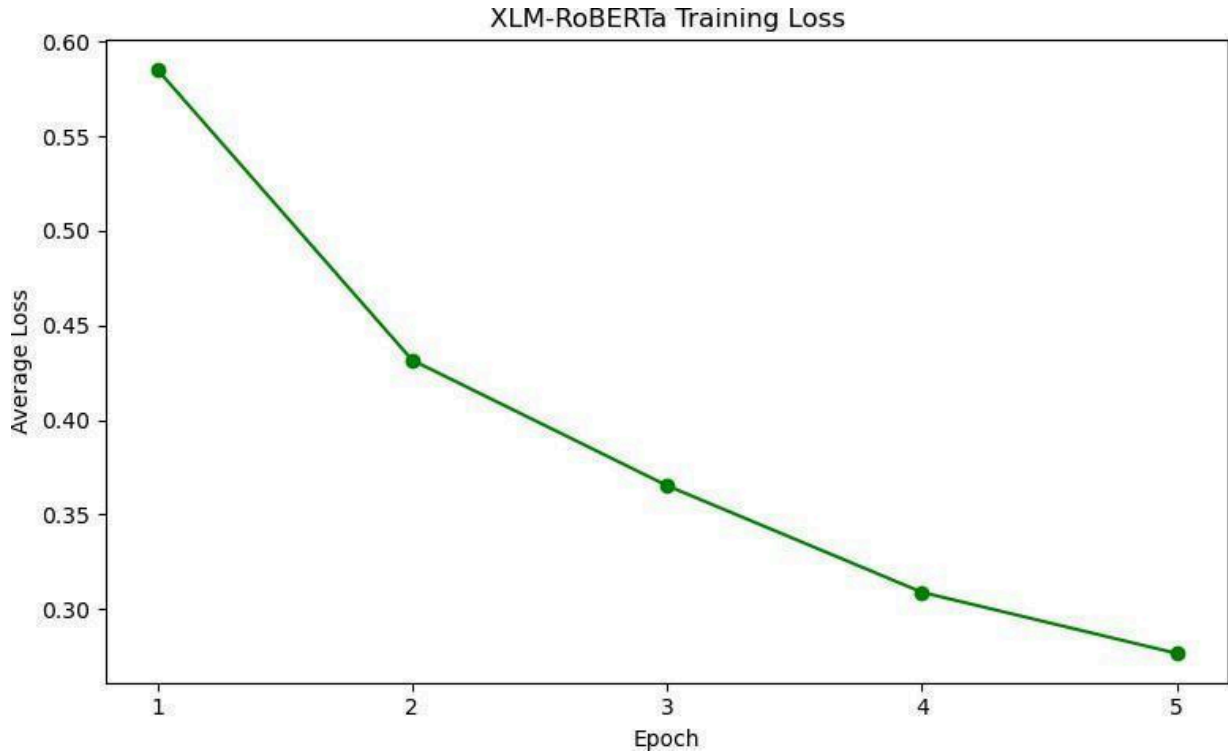


Figure 5: XLM-RoBERTa training loss across five fine-tuning epochs, demonstrating consistent convergence without overfitting.

4.4 Out-of-Domain Stress Test

XLM-RoBERTa scored 81% accuracy when applied directly to the 8,147-instance emotion corpus without any further fine-tuning; this is exactly one percentage point less than its in-domain performance on RUSAD. Table 4 displays the whole classification report.

Class	Precision	Recall	F1-Score	Support
Negative	0.86	0.79	0.82	4,427
Positive	0.77	0.84	0.80	3,720
Weighted Average	0.82	0.81	0.81	8,147

Table 4: XLM-RoBERTa classification report on the out-of-domain stress test.

Figure 7's confusion matrix reveals that 3,135 positive cases were correctly identified and 585 incorrectly forecasted as negative, while 3,486 negative instances were correctly classified and 941 incorrectly anticipated as positive. The precision ratings for the two classes show a clear asymmetry (0.86 for negative and 0.77 for positive), suggesting that the model is much more cautious when predicting negative sentiment on the emotion corpus than it was on RUSAD. This could be explained by the fact that the emotion dataset's negative class is made up of three

different emotion categories: anger, sadness, and fear. Each of these categories has a unique surface language profile, which could lessen the consistency of the negative signal the model receives.

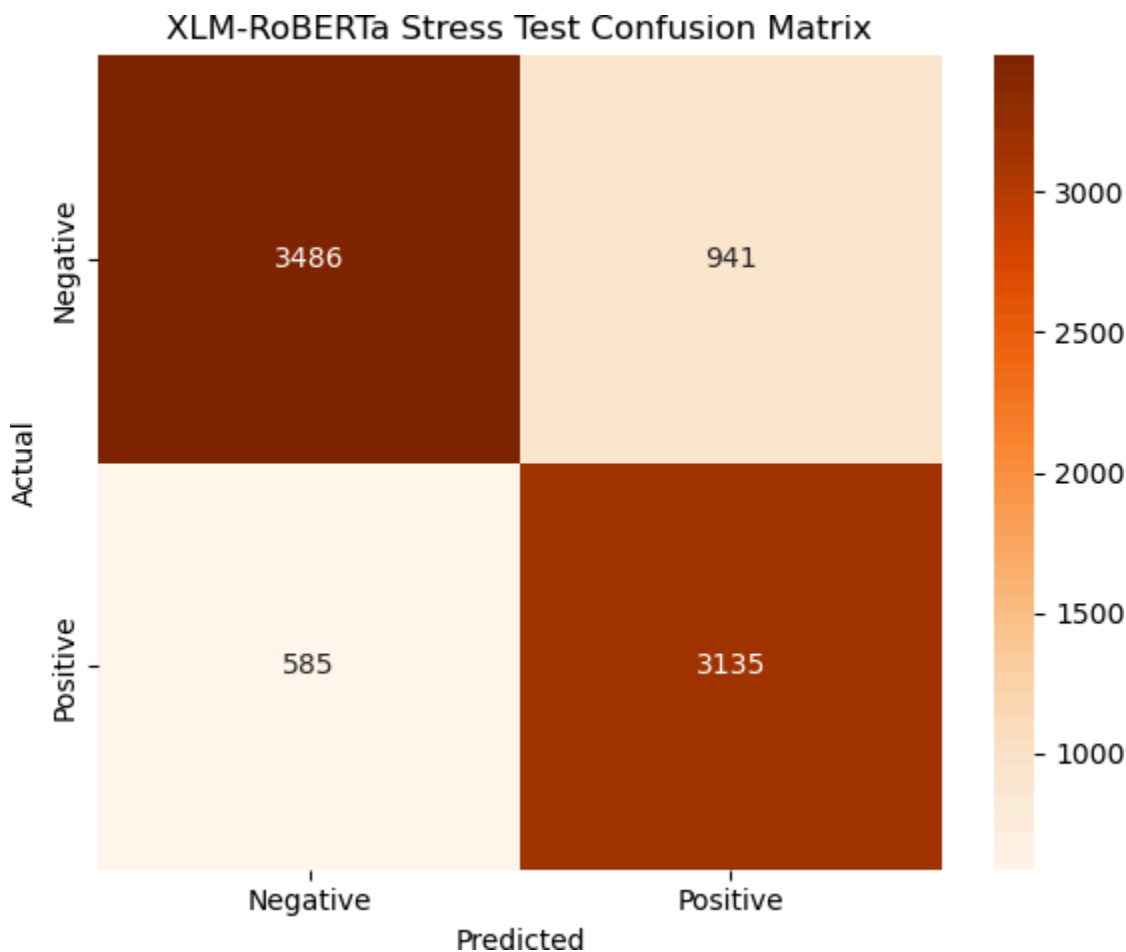


Figure 6: Confusion matrix for XLM-RoBERTa evaluated on the out-of-domain Roman Urdu-English Code-Switched Emotion Dataset.

4.5 Error Analysis

1,526 of the 8,147 stress test cases were incorrectly classified, resulting in an overall error rate of 18.7%. These mistakes are not distributed randomly. When the misclassified cases are examined, they show a strong concentration in the Anger emotion category, and they all have a surface feature that is readily noticeable upon closer scrutiny. A sample of misclassified cases is shown in Table 5.

Text	Emotion	Actual	Predicted
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Ok Bhai theek hai	Anger	Negative	Positive
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Night duty pai hai	Anger	Negative	Positive
Pakistan me CNIC pe Money Transfer ki option hai	Anger	Negative	Positive
Tera store khula hai, inhe soup pilane ko	Anger	Negative	Positive

Table 5: Representative misclassified instances from the stress test error analysis.

The total lack of terminology that is overtly negative is what brings these cases together. A native speaker of Roman Urdu social media discourse would instantly recognise the tone of frustrated resignation in the phrase "Ok Bhai theek hai," which roughly translates to "Ok fine whatever" in English. However, its surface tokens lack a polarity signal that a model trained on binary sentiment labels would be able to identify. Similarly, rather than using any explicitly negative vocabulary meaning, "Night duty pai hai"—"I am on night duty"—communicates displeasure by situational implication. The model, having been trained on RUSAD reviews in which sentiment is expressed more explicitly, classifies these instances as positive on the basis of their surface features.

SECTION 5 — DISCUSSION

The findings in Section 4 raise a number of issues that should be carefully considered. It is correct that XLM-RoBERTa outperforms Bi-LSTM by two percentage points on the RUSAD holdout set, as reported in the headline result. However, the reason why the difference isn't larger is more interesting. This section discusses both of those problems as well as what the stress test and error analysis reveal about the more widespread challenges with Roman Urdu sentiment analysis.

It would be easy to conclude that XLM-RoBERTa is superior based only on the performance differential between the two versions. However, without comprehending the underlying mechanism, that explanation is not very helpful. The out-of-vocabulary issue is the biggest structural challenge for Roman Urdu sentiment analysis, as noted in Section 2. Because Roman Urdu lacks standard spelling, any model that relies on a fixed vocabulary will frequently come across word forms it has never seen before.

with XLM-RoBERTa correctly classifying 59 more cases and lowering the false negative rate from 21.3% to 16.1%. Positive sentiment in Roman Urdu social media is frequently expressed through enthusiasm, colloquial praise, and culturally specific expressions of approval that are more likely to involve code-mixing or spelling variation than straightforwardly negative reviews, which typically use more direct and recognisable negative vocabulary. This could be one explanation for this asymmetry.

When comparing these results to previous research, XLM-RoBERTa's 82% accuracy is less than Li et al. (2022)'s 90.3% accuracy on the identical RUSAD dataset. Instead of viewing this disparity as a flaw, it is crucial to contextualise it. Four embedding channels, attention mechanisms, and multi-domain transfer learning make up Li et al.'s much more complicated design, which takes a lot more technical work and processing power to set up and fine-tune. This study used a conventional fine-tuning procedure that would be easy to repeat, with a deliberate focus on reproducibility and simplicity.

Examination, however, reveals a flaw that needs serious attention. The Anger group comprises the majority of the 1,526 improperly classified test cases, and they all share the trait of not using any obviously negative terminology. In Roman Urdu social media culture, phrases like "Ok Bhai theek hai" carry sarcastic or annoyed implications that are completely undetectable at the surface linguistic level, as was discussed in Section 4. The issue is not with the model's ability to recognise negative words, but rather with its ability to recognise sentiment that is expressed indirectly through cultural context, implication, and pragmatic meaning as opposed to explicit polarity signals. This issue can be easily fixed with better architecture or additional training data.

SECTION 6 — CONCLUSION

The lack of a direct, controlled comparison between sequential deep learning and multilingual transformer architectures on the RUSAD benchmark and the absence of any systematic evaluation of cross-domain generalisation for this task were the specific and practically significant gaps in Roman Urdu natural language processing that this study aimed to fill.

The results show clearly that fine-tuned XLM-RoBERTa outperforms a Bidirectional LSTM baseline on binary Roman Urdu sentiment classification, achieving 82% accuracy and a weighted F1-score of 0.82 compared to 80% and 0.80 for Bi-LSTM. The primary mechanism behind this improvement is XLM-RoBERTa's subword tokenisation, which handles the orthographic inconsistency endemic to Roman Urdu by decomposing unknown spelling variants into recognisable character-level fragments rather than discarding them as unknown tokens. The out-of-domain stress test further showed that the fine-tuned model generalises with notable robustness, maintaining 81% accuracy on a structurally different emotion dataset it had never encountered during training.

Irony, sarcasm, and culturally specialised indirect language that lacks clear negative terminology are examples of the systematic misclassification of sentiment. This suggests a path for future research that may be more significant than additional architectural improvement: the creation of training corpora and annotation frameworks that capture both surface polarity and pragmatic, culturally grounded emotion. The field's ability to simulate the genuine complexity of this language as it is really used would be greatly advanced by a bigger, neutrally-labeled Roman Urdu corpus covering the entire tripartite sentiment space—positive, negative, and neutral.

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